

Drishti

Karnataka holds thousands of undetected cases that stay isolated for a structural reason: the FIR schema carries **no global offender identity**. `Accused.PersonID` is documented as "A1, A2, A3..." — a sort order inside a single FIR. When one offender works across several stations the database holds several unrelated name strings, and nobody joins them. **Drishti links those cases by how the crime was committed rather than by who was named**, and returns a ranked, evidence-backed queue of offender groups to reopen.

789	16	16	87	0.946	13/13
UNDETECTED CASES	OFFENDER GROUPS	CROSS-STATION	CASES CONNECTED	F1 SCORE	GROUPS RECOVERED

HOW IT WORKS

Every undetected FIR gets a **behavioural fingerprint** built from seven weighted signals: the modus operandi in the complaint text (TF-IDF), crime sub-head, the Act and Section invoked, location, time-of-day and day-of-week pattern, victim profile, and accused name. Composite similarity is computed for every pair of cases, with **weights renormalised per pair** so a case with sparse free text still links on typology, place and time rather than dropping out. The resulting distance matrix is clustered with agglomerative linkage, and clusters are ranked by cohesion × actionability — which favours jurisdictional reach, recency and gravity, because a group inside one station is a lead that station already has.

WHAT THE PROTOTYPE DOES

- **Case linkage** — ranked offender groups from the undetected pool, every one citing real FIR numbers
- **Evidence trail** — per-signal contribution, spatial span and full timeline for each group
- **Next-strike projection** — window, area radius and usual hours from each group's own rhythm and patch
- **Live FIR triage** — an incoming case scored against every group, graded against that group's own cohesion
- **Persons of interest** — 48 names recurring across separate FIRs, 47 of them across station boundaries
- **3D district map** — WebGL terrain extruded from real boundaries, with case pins, link webs and forecast rings
- **Role-based access** — investigator / analyst / supervisor, enforced at the gateway
- **Case brief PDF**, bilingual English / Kannada interface, and an in-product accuracy panel

MEASURED PERFORMANCE

LINKAGE ACCURACY		SIGNAL REMOVED	F1	COST
Precision	0.928	Location	0.363	-0.583
Recall	0.965	Method (MO text)	0.438	-0.508
F1	0.946	Crime type	0.737	-0.209
Planted groups recovered	13 of 13	Who was targeted	0.780	-0.166

Pairwise over every pair of undetected cases against planted ground truth hidden from the engine. Reproducible offline:

```
python appsail-python/run_local.py
```

Leave-one-signal-out ablation: each weight is zeroed and the whole pipeline re-run, so the drop is what that signal is actually worth.

THE IDENTITY EXPERIMENT

We assumed the accused name would help link cases and tested it. It made the engine **worse** — with the name at 8% weight, precision 0.919 / recall 0.854 / F1 0.885; at zero weight, **0.928 / 0.965 / 0.946**. Coincidental name collisions were pulling unrelated cases into real groups. Identity therefore carries **no weight in linkage**; a recurring name is surfaced to the investigator as unconfirmed corroboration only. Reporting delay and complainant occupation were built and measured the same way, and also ship at zero — the synthetic corpus draws both at random.

Forecast, back-tested. Each group's last offence was held out, a projection built from the earlier events only, then checked against what actually happened: **area hit rate 92.9%**, window hit rate 50.0%, median timing error 7.7 days, across 14 groups. The area holds far better than the timing, which is why the map ring is the actionable part of the projection rather than the date.

ARCHITECTURE AND CATALYST SERVICES

SERVICE	ROLE IN THE SYSTEM
AppSail	Flask ML engine — fingerprinting, clustering, forecast, triage, PDF rendering
Serverless Functions	API gateway; resolves role and enforces access before any handler reads case data
Data Store	FIR schema — 28 tables, queried by ZCQL
Authentication	Identity behind the three roles
Cache	Computed API responses shared across function instances
Job Scheduling	Nightly re-cluster and cache flush — used in place of Cron, which is deprecated
Signals + Event Functions	Auto-triages a newly registered FIR — in place of Event Listeners, also deprecated
SmartBrowz · Web Client Hosting	Case-brief PDF rendering · serves the React application

Stack. React 19, TypeScript and Three.js on the client; Python 3.13, Flask and scikit-learn on the engine. Clustering stays on scikit-learn primitives deliberately, so the engine installs from `requirements.txt` on the managed runtime with no native compilation. **No GPU and no external LLM API** — retrieval is deterministic and case data never leaves the Catalyst environment.

CURRENT STATE

Live and working: linkage, evidence trails, forecasting, live triage, persons of interest, access control, PDF export, the accuracy panel and the 3D map — all deployed and reachable at the link below. **Schema conformance is exact:** 28 of 28 ER tables implemented, 42 of 42 foreign keys resolve, and the engine reads 25 distinct columns, every one of them present in the ER diagram.

CONSOLE-GATED, NOT MISSING

Four items need an admin console action that neither the CLI nor the SDK exposes: **Data Store table creation** (the engine has a working Data Store backend and runs on bundled schema-faithful CSVs until the tables exist, which also enables full-text search on **BriefFacts**), a **Production environment**, a **jobpool** to schedule the deployed recompute job, and the **Signals binding** for the deployed event function.

LIMITS, STATED DELIBERATELY

- Every link is **inferred, never asserted** — a direction for investigation, not evidence.
- Two unrelated offenders working the same market at the same hour can look alike; spread-out groups are damped in ranking, but a tight cluster is not proof.
- Sparse complaint text weakens the strongest signal. Weights renormalise so linkage still runs, and the cohesion score reflects the reduced confidence.
- Only **233 of 789** cases name an accused at all, which is precisely why name recurrence is a lead rather than a conclusion.
- Complainant caste and religion exist in the schema and are **deliberately excluded** from linkage; only victim age and gender inform the target profile.
- Figures are measured on schema-faithful synthetic data, since the KSP dataset was not released. Real FIR data swaps in through the same tables and pipeline unchanged, and the accuracy panel re-measures against it.

Deployed `crimelink-927199644.development.catalystserverless.com/app/index.html`

Repository `github.com/whysush/drishti-crimelink`

Worth opening `?tab=model` (measured accuracy) · `?group=SER-001` (a group with its forecast) · `?role=investigator&district=577` (access control)

Every figure in this brief was read live from the deployed /validation endpoint. The engine idles between uses — open the link a minute early and wait for ENGINE LIVE.